

# I-BEAMS

| No. | Load condition | Bending moment                                                                                                                                                                | Reaction force and shear stress                                                                                                                 | Deflection                                                                                           |
|-----|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| 21  |                | $M_A = -\frac{Pab^2}{l^2}$ $M_B = -\frac{Pa^2b}{l^2}$ $M_C = \frac{2Pa^2b^2}{l^3}$                                                                                            | $R_A = P \left( \frac{b}{l} \right)^2 \left( 1 + 2 \frac{a}{l} \right)$ $R_B = P \left( \frac{a}{l} \right)^2 \left( 1 + 2 \frac{b}{l} \right)$ | $d_{max} = \frac{Pa^3b^3}{3EI l^3}$ $d_{max} = \frac{2Pa^2b^3}{3EI(3l-2a)} \times \frac{l^2}{3l-2a}$ |
| 22  |                | $M_A = M_B = -2Pl/9$ $M_C = M_D = Pl/9$                                                                                                                                       | $R_A = R_B = P$                                                                                                                                 | $d_{max} = \frac{5Pl^3}{648EI}$                                                                      |
| 23  |                | $M_A = M_B = -19Pl/72$ $M_D = 11Pl/72$                                                                                                                                        | $R_A = R_B = 3P/2$                                                                                                                              | $d_{max} = \frac{41Pl^3}{5184EI}$                                                                    |
| 24  |                | $M_A = M_B = -5Pl/16$ $M_D = 3Pl/16$                                                                                                                                          | $R_A = R_B = 3P/2$                                                                                                                              | $d_{max} = \frac{Pl^3}{96EI}$                                                                        |
| 25  |                | $M_A = M_B = -\frac{Wl^2}{12}$ $M_C = \frac{Wl^2}{24}$                                                                                                                        | $R_A = R_B = W/2$                                                                                                                               | $d_{max} = \frac{Wl^3}{384EI}$                                                                       |
| 26  |                | $M_A = -\frac{W}{12Pl^2} [e^3(4l-3e) - c^3(4l-3c)]$ $M_B = -\frac{W}{12Pl^2} [d^3(4l-3d) - a^3(4l-3a)]$                                                                       | $R_A = r_A + \frac{M_A - M_B}{l}$ $R_B = r_B + \frac{M_B - M_A}{l}$ <p><math>r</math> indicates the reaction force of the single span.</p>      | $d_{max} = \frac{W}{384EI} (l^3 + 2l^2a + 4la^2 - 8a^3)$                                             |
| 27  |                | $M_X = -\frac{Wl}{30} \left( \frac{10x^3}{l^3} - \frac{9x}{l} + 2 \right)$ <p>where <math>+M_{max} = Wl/23.3</math>, <math>x = 0.55l</math></p> $M_A = -Wl/15$ $M_B = -Wl/10$ | $R_A = 0.3W$ $R_B = 0.7W$                                                                                                                       | $d_{max} = \frac{Wl^3}{382EI}$ <p>where <math>x_1 = 0.525l</math></p>                                |
| 28  |                | $M_A = M_B = -\frac{5Wl}{48}$ $M_C = Wl/16$                                                                                                                                   | $R_A = R_B = W/2$                                                                                                                               | $d_{max} = \frac{1.4Wl^3}{384EI}$                                                                    |
| 29  |                | $M_A = M_B = -\frac{W}{48l} x$ <p><math>(5l^2 + 4al - 4a^2)</math></p>                                                                                                        | $R_A = R_B = W/2$                                                                                                                               | $d_{max} = \frac{W}{1920EI} (7l^3 + 8a^2l + 4a^2l - 16a^3)$                                          |
| 30  |                | $M_A = M_B = -\frac{Wa}{12l} (2l - a)$                                                                                                                                        | $R_A = R_B = W/2$                                                                                                                               | $d_{max} = \frac{Wa^2}{480EI} (5l - 4a)$                                                             |

In case of beam subject to distributed loads, load symbol, W, indicates the total load of the beam. For example, in case of a beam subject to uniformly distributed load, w, W equals  $w\ell$ , where  $\ell$  is span length.